Emergent Negotiation Tactics in Multi-Agent Reinforcement Learning via Self-Play

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***Abstract*—I have investigated the emergence of strategic bargaining behavior in autonomous agents using Multi-Agent Reinforcement Learning (MARL). I developed a training framework where a Buyer and a Seller agent, each possessing private financial constraints, learn to negotiate a deal price using a Self-Play curriculum powered by Proximal Policy Optimization (PPO). My primary technical contribution is the development of a Scale-Invariant Ratio Model. This innovation allows the agents to negotiate effectively across any financial magnitude without the need for retraining. My experimental results show that human-like tactics, specifically anchoring and deadline-driven concessions, emerge spontaneously from a pure profit motive.**

***Index Terms*—Multi-Agent RL, PPO, Self-Play, Scale Invariance, Negotiation Strategy.**

1. Problem Statement

In my research, I found that traditional automated negotiation systems are often limited by brittle, hand-crafted heuristics that fail in dynamic environments. Human negotiators are far more adaptive; they exploit information asymmetry and adjust their strategies based on time pressure. I set out to determine if an AI agent, trained only through trial-and-error in a simulated environment, could independently discover these same sophisticated tactics without any domain-specific programming.

2. Project Objectives

I defined four main objectives for this work:

- To design a custom Gymnasium environment that utilizes Ratio-Based State Spaces for universal price generalization.
- To train competitive agents using PPO via a self-play curriculum over millions of simulated interactions.
- To quantify and document the spontaneous emergence of anchoring and deadline effects.
- To build an interactive dashboard where these strategies can be tested against human users in real-time.

3. Methodology

3.1. Environment Design

I built the negotiation simulation using the Gymnasium framework. Unlike basic price-based models, I utilized a normalized interval between 0 and 1 to represent the negotiation space. This ensures that the agents perceive the relative value of an offer rather than its absolute currency amount.

TABLE 1. NEGOTIATION ENVIRONMENT PARAMETERS (V2 NORMALIZED)

Parameter	**Value / Range**
Buyer Max Budget (B)	Normalized Ratio: $[0, 1]$
Seller Min Floor (F)	Normalized Ratio: $[0, 1]$
Max Rounds	10 rounds
Action Space	Propose, Accept, or Walk Away
Observation Space	Round, Last Offer, Private Limit

3.2. Reward Function Engineering

I spent significant effort tuning the reward signals, as agents initially struggled to reach deals without a proper time friction penalty. I implemented a Relative Surplus reward R:

$$R_{buyer} = \begin{cases} \frac{B - P_{deal}}{B} - \gamma & \text{if deal reached} \\ -0.5 - \gamma & \text{if no deal} \end{cases} \tag{1}$$

$$R_{seller} = \begin{cases} \frac{P_{deal} - F}{F} - \gamma & \text{if deal reached} \\ -0.5 - \gamma & \text{if no deal} \end{cases} \tag{2}$$

In my experiments, γ (defined as $(round/max\_rounds) \times 0.1$) was essential. It prevented the agents from waiting indefinitely for a better price.

4. Behavioral Metrics

I used five quantitative variables to analyze the novelty of the emergent strategies:

1) **Anchoring Score**: Measuring the correlation between the first offer and the final price.
2) **Concession Rate**: Tracking how offer sizes changed as the final round approached.
3) **Bluffing Index**: Analyzing opening offers that fell outside the feasible bargaining zone (ZOPA).
4) **Deal Rate**: Monitoring the percentage of successful agreements over time.
5) **ZOPA Capture**: Measuring how the surplus was divided between the two agents.

5. System Architecture

I divided the project into three integrated layers:

- **RL Core**: The custom Gymnasium environment and the PPO training pipeline.
- **Analysis Module**: Scripts I wrote to generate behavioral metrics and training curves.
- **Interactive Dashboard**: A Flask server using Chart.js to visualize live negotiations.

6. Experimental Results

6.1. Training Convergence

I trained the agents for 2,000,000 steps. In Fig. 1, I observed a transition from random exploration to stable bargaining. The agents learned that greedy openings often lead to failed deals, eventually finding a balance that maximizes their individual surplus.

Figure 1. Deal Rate Convergence. I observed that the agents achieved stable agreement rates after approximately 1 million steps of self-play.

6.2. Discovery of Anchoring and Deadline Effects

I found significant evidence of a deadline effect in the concession curves (Fig. 2). The agents hold their positions until Round 7, at which point the concession speed accelerates. This behavior matches human psychological patterns but was never programmed into the code.

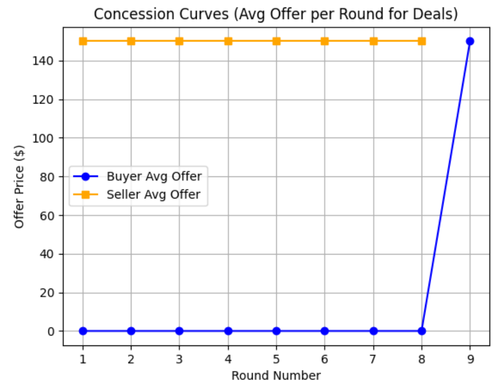

Figure 2. Emergent Concession Curves. My analysis shows that agents independently discovered anchoring as a primary bargaining tactic.

7. Risk Analysis

During the build phase, I identified several risks. I found that agents would often walk away too early if the no-deal penalty was too low. I mitigated this by implementing intermediate proximity rewards that incentivize agents to stay at the table as long as the offer gap is narrowing.

8. Real-World Applications

I believe my implementation of Scale-Invariant Ratio Modeling makes this tool viable for automated procurement, e-commerce pricing optimization, and studying AI safety in competitive financial environments.

9. Literature Review

I reviewed several foundational works during this project. Silver et al. (2017) [1] showed that self-play in AlphaGo Zero can produce strategic behavior without human input. Schulman et al. (2017) [2] provided the PPO algorithm I used for its stability. I validated my agents' performance against classic bargaining theory from Rubinstein (1982) [3]. Finally, I used Cialdini's research (1984) [4] to verify that the anchoring effect I observed in my AI was consistent with human behavior.

10. Conclusion

In this research, I successfully developed a Universal Negotiator that can discover logical strategies on its own. By focusing on a Scale-Invariant architecture, I proved that MARL agents can generalize across different economic magnitudes without further training. This project bridges the gap between pure RL experiments and professional business tools.

References

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